

# Resume

## Bui Xuan Toan



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## Educational Record

- **2007–2009:** Le Quy Don High School, Phan Rang – Thap Cham City, Ninh Thuan Province
- **2009–2013:** Bachelor's Degree in Finance-Banking, Ho Chi Minh City Open University
- **2015–2017:** Master's Degree in Industrial Management, Graduate Institute of Industrial Management, National Central University, Taiwan

## Work Experience

- **2013–2015:** Financial Analysis – Creditor (Personal Customer) at VietinBank Ninh Thuan, Ninh Thuan Province
- **07/2017–08/2018:** Planning Function Developer at Fusheng Precision, Taoyuan, Taiwan
- **08/2018–10/2020:** Business Process Automation at Grab (Regional Team)
- **11/2020–08/2021:** Data Engineer at Just Analytics
- **09/2021–12/2023:** Data Engineer (Project Lead) at HCL
- **01/2024–06/2024:** Data Engineer / Solution Architect (Independent Contractor) at HailstoneLab
- **07/2024–02/2025:** Senior Data Engineer (Contractor) at Grab (Data Ingestion Team)
- **02/2025–present:** Senior Data Engineer at Persefoni

## Certificates

- TOEIC: 805 (2011)
- IELTS: Listening 8.0 | Reading 7.0 | Writing 6.0 | Speaking 6.0 | Overall 7.0 (2014)
- Google Cloud Data Engineer (2019)
- Google Cloud Data Architect (2020)
- Google Cloud DevOps Engineer (2023)
- Google Cloud Security Engineer (2023)
- AWS Certified DevOps Engineer – Professional (2024)
- AWS Certified Solutions Architect – Professional (2024)
- Databricks Certified Data Engineer Professional (2024)

- SnowPro Core Certification (2025)
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## Skills

### Core CS & Software Engineering

- Solid understanding of computer science concepts, statistics, and software development
- Data structures, classic algorithms, and complexity analysis
- Linux administration, containerization (Docker), container orchestration (Kubernetes)
- Process, thread, and async programming
- HTTP protocol, REST API design
- Strong advocate of **software quality** — apply **Test-Driven Development (TDD)** as a core practice: write tests first, drive design through failing tests, and refactor with confidence
- Practice **Continuous Integration / Continuous Deployment (CI/CD)**: automate build, test, and deployment pipelines to catch issues early and ship reliably
- Apply **Clean Code principles** (Robert C. Martin) daily: meaningful naming, small focused functions, single responsibility, minimal side effects, and keeping code readable as a first-class concern
- Apply **Clean Architecture** to separate business logic from infrastructure, making systems easier to test, maintain, and evolve

### Python

- Fluent in Python with strong knowledge of built-in data structures
- Object-oriented design and implementation
- Support libraries for cloud services (AWS, GCP)
- Performance improvement using `asyncio` and `httpx`
- Test-driven development and code quality focus

### Data Analytics

- Data wrangling with pandas, numpy, and Jupyter Notebook
- Data visualization with matplotlib, seaborn, and plotly
- Data preparation and cleaning for optimization models

### Data Engineering & Data Warehousing

- Advanced SQL query writing and optimization across relational databases
- Experience with traditional databases: PostgreSQL, MySQL, SQL Server
- Experience with columnar/cloud databases: BigQuery, Presto (Trino), Databricks Delta Tables, Snowflake
- Distributed data processing: Apache Spark, Apache Beam, Apache Flink
- **Data warehouse design** focusing on scalable architectures (Star/Kimball schema design, Medallion architecture)
- **Data Reliability & Testing**: Implement robust testing frameworks (dbt tests, custom validations) to ensure **data correctness** and **freshness**; apply TDD to data modeling
- **Observability**: Implement end-to-end monitoring for data platforms, tracking pipeline health, data quality, and lineage; **leverage query tagging** (e.g., Snowflake query tags) to perform granular analysis of query execution time and cost across complex data models
- **Cost Optimization**: Optimize cloud warehouse spend (Snowflake, BigQuery) through query tuning, resource management, and storage optimization (e.g., achieved 33% cost reduction)
- Orchestration with Apache Airflow
- Change Data Capture (CDC) with Debezium, Kafka Connect, Kafka Server
- Kafka producer/consumer using confluent-kafka Python client

- Data warehouse design and testing with dbt (test-driven approach)

Visualization

- Apache Superset, Looker Studio, Tableau, Power BI
- CI/CD design for visualization platforms (version control, safe deployment)
- Cost optimization with caching and logical data model tuning
- Multi-tenant access control setup

Backend & AI

- REST API development with Flask, Django, FastAPI
- LLM agent development using LangChain, LangGraph, LangServe

Vibe Coding (AI-Assisted Software Development)

- Proficient in AI-assisted development using tools such as Claude Code, GitHub Copilot, and Cursor to accelerate feature delivery without sacrificing quality
- Apply **Test-Driven Development (TDD)** as the anchor of AI-assisted workflows: write failing tests first, use AI to generate implementation, then refactor — ensuring generated code is always validated against a clear specification
- Treat AI-generated code with the same rigor as human-written code: review for correctness, security, edge cases, and maintainability before accepting
- Use AI tools to scaffold boilerplate, suggest refactors, and explore approaches, while retaining full ownership of design decisions and code quality
- Enforce code quality gates (linting, type checking, test coverage thresholds) in CI/CD pipelines to catch issues in AI-generated code automatically
- Focus on prompt engineering for code generation: provide context, constraints, and expected behavior to guide AI toward correct and minimal solutions

Research Experience

- Applied Genetic Algorithm to Schedule Brewery Production (*Master's thesis*)
- Breast Cancer Prediction (*Machine Learning coursework*)
- American Presidential Election Prediction (*NLP coursework – correctly predicted Trump's win*)

Highlight Projects

| Project                                      | Company           | Scope           | Result                                                                            |
|----------------------------------------------|-------------------|-----------------|-----------------------------------------------------------------------------------|
| Compute WIP                                  | Fusheng Precision | Fusheng Vietnam | Reduced WIP report generation time from hours to seconds                          |
| Highlight Late Kanban Cards                  | Fusheng Precision | Fusheng Vietnam | Helped managers identify late cards for further investigation                     |
| Compute Incentive for Grab Express           | Grab              | Grab Vietnam    | Replaced manual Excel computation. Stack: Python, SQL, pandas, Athena, Google API |
| Automation Send Email for Drivers' Incidents | Grab              | Grab Vietnam    | Eliminated manual email writing. Stack: Python, SQL, pandas, Athena, AWS SES      |

|                                              |                |                      |                                                                                                                                       |
|----------------------------------------------|----------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Data Analysis for QC Team                    | Grab           | Grab Vietnam         | Cleaned database for better driver management. Stack: Python, MySQL, pandas, Athena, Google API                                       |
| Collect Drivers' Commission Fee for Partners | Grab           | Grab Indonesia       | Faster and more accurate calculations. Stack: Python, MySQL, pandas, Athena, Google API                                               |
| Grab Hitch                                   | Grab           | Grab Hitch Singapore | Supported Hitch team's core calculations (points, tier, commission) for 100K+ users. Stack: Python, MySQL, pandas, Athena, Google API |
| Mex Correction                               | Grab           | Grab Food Regional   | Summarized driver restaurant incident reports to reduce ticket volume. Stack: Python, MySQL, pandas, Athena, Google API               |
| Mex Insight                                  | Grab           | Grab Food Regional   | Automated delivery report process for restaurant partners. Stack: Python, MySQL, pandas, Athena, Google API                           |
| Mex Billing                                  | Grab           | Grab Food Regional   | Built billing system for Grab Food. Stack: Python, MySQL, pandas, Athena, Google API                                                  |
| Dax Event Tracker                            | Grab           | Grab Regional        | Automated event invitations, registration, and analysis. Stack: Python, MySQL, pandas, Athena, Google API                             |
| Data Lake                                    | Just Analytics | Nam Long Corporation | Custom data pipeline for finance and assets data. Stack: Python, SQL, SQL Server                                                      |
| Data Lake Migration                          | Just Analytics | Grab                 | Migrated data pipeline from AWS to Azure. Stack: Python, SQL, Azure Databricks                                                        |
| KOL Database                                 | HCL            | Amorepacific         | Deployed Airflow with Docker, built pipeline for YouTube channel/video data. Stack: Python, SQL, web scraping, AWS RDS MySQL          |
| Mall Review Database                         | HCL            | Amorepacific         | Built web crawling pipeline for product reviews. Stack: Python, SQL, web scraping, AWS RDS MySQL                                      |
| Maintain Batch Data Pipeline                 | HCL            | ANZ                  | Automated metadata creation, query partition optimization. Stack: Python, SQL, BigQuery, Airflow, dbt                                 |
| Group Management System                      | HCL            | ANZ                  | GCP IAM-based group management. Stack: Git, Terraform, Bash                                                                           |

|                                          |            |               |                                                                                                                                                                               |
|------------------------------------------|------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Deploy Datahub to Shared Cluster         | HCL        | ANZ           | Deployed Datahub to GKE shared cluster. Stack: Helm, Kubernetes, Bash                                                                                                         |
| Build em_helper Python Package           | HCL        | ANZ           | Support package for Vault and Kafka communication. Stack: Python, Kafka                                                                                                       |
| Infrastructure as Code                   | Contractor | HailstoneLabs | Centralized GCP resource provisioning for dev and prod. Stack: GitHub, Terraform, Cloud Build                                                                                 |
| CI/CD for Airflow                        | Contractor | HailstoneLabs | Built CI/CD to build and deploy Airflow. Stack: Helm, GKE, GCP Artifact Registry, Cloud Build, Docker                                                                         |
| Role-Based Access Management             | Contractor | HailstoneLabs | Implemented and transferred knowledge on RBAC                                                                                                                                 |
| Twitter Dashboard Data Pipeline          | Contractor | HailstoneLabs | Built pipeline to track VC activity on Twitter. Stack: Python, SQL, Airflow, web scraping, dbt, BigQuery. Saved ~\$40,000/month by replacing Enterprise API with web scraping |
| Crypto Market Tracking Pipeline          | Contractor | HailstoneLabs | Dashboard tracking crypto market movements. Stack: Python, SQL, Airflow, dbt, BigQuery, Slack app                                                                             |
| ETH Block Data Pipeline                  | Contractor | HailstoneLabs | Ingested Ethereum node data, optimized for queries. Stack: Python, SQL, Airflow, dbt, BigQuery                                                                                |
| Hugo LLM Agents for Auto-Healing         | Contractor | Grab          | Built and deployed LLM agents for automated data pipeline debugging. Stack: LangChain, LangGraph, LangServe                                                                   |
| Airflow 1 → Airflow 2 Migration          | Contractor | Grab          | Migrated pipelines to modern Python-based Airflow with improved secrets management                                                                                            |
| Cross-Region Data Copy (SG → VN)         | Contractor | Grab          | Copied large volumes of data from Singapore to Vietnam AWS for Decree 53 compliance                                                                                           |
| OVO Data Warehouse Migration (GCP → AWS) | Contractor | Grab          | Migrated OVO Indonesia data pipelines and warehouse from GCP to AWS                                                                                                           |
| Scope 1/2/3 Emission Data Model          | Persefoni  | Persefoni     | Fixed long-standing bugs causing incorrect calculations; applied TDD for stable releases. Stack: Debezium, Kafka, Snowflake, dbt, GoodData                                    |
| Waste & Water Data Pipeline              | Persefoni  | Persefoni     | Delivered bug-free data model and visualization through thorough testing                                                                                                      |

|                                               |           |           |                                                                                                          |
|-----------------------------------------------|-----------|-----------|----------------------------------------------------------------------------------------------------------|
| Financial Data Model (France & Other Markets) | Persefoni | Persefoni | Extended model for multi-language/market support; fixed hidden bugs                                      |
| Databricks ETL → CDC Migration                | Persefoni | Persefoni | Improved data freshness using CDC and Snowflake streaming architecture                                   |
| Snowflake Cost Optimization                   | Persefoni | Persefoni | Reduced Snowflake costs by ~33%                                                                          |
| Emission & Energy Use Intensity Model         | Persefoni | Persefoni | Built model supporting facility occupied area changes over time and complex intensity metric computation |